

Moral Tensors and DecisionProofs

Compiling Language into an Auditable, Grounded Safety Layer

Andrew H. Bond
San José State University & independent researcher
ORCID 0009-0003-2599-6158

PyTorch Conference North America 2026, Responsible AI poster

Most AI safety is a scalar at the output: one good/bad check before an action ships. We keep the structure instead (who is affected, what is owed, who consented, who bears imposed risk) and contract to a decision only at the end, auditably.

Two open-source pieces do this. `erisml-compiler` turns natural-language moral material into a typed MoralGraph and a moral tensor, read through four ethical lenses at once. When the lenses disagree it does not aggregate them silently. It defers to a human. At runtime, `erisml-lib` puts the same evaluation into an agent's plan→act loop, seals each decision in a hash-chained DecisionProof, and fails safe, never open.

1. Structure before contraction: keep the tensor, log the collapse and the residue.
2. Pluralism is the responsible move: four lenses, and a human on conflict.
3. Auditable by construction: fails safe, never open.

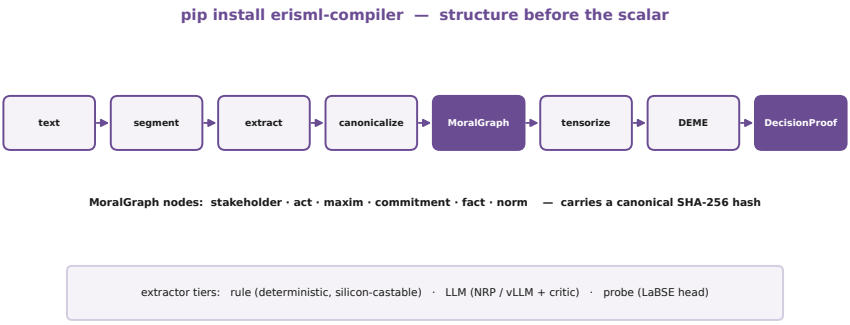
The full companion paper (4 pages)

Structure before the scalar

A single good/bad score discards what an audit needs: which stakeholders are affected, what is owed to each, who consented, and who bears imposed risk. ErisML keeps that structure as a typed object through the whole evaluation and performs *decision contraction* only at the end, logging the weights used, who lost, and what moral residue remains. The claim is not that the system knows what is good. It is that every collapse from structure to verdict is recorded and replayable.

The compiler

`erisml-compiler` (alpha v0.9.0, PyPI, MIT, 477 tests) runs a 12-pass pipeline. In outline:



The MoralGraph is typed (stakeholder, act, maxim, commitment, fact and norm nodes) and carries a canonical SHA-256 hash. Extraction has three tiers: deterministic rules, an LLM tier (vLLM with a critic pass), and a LaBSE classifier-head probe.

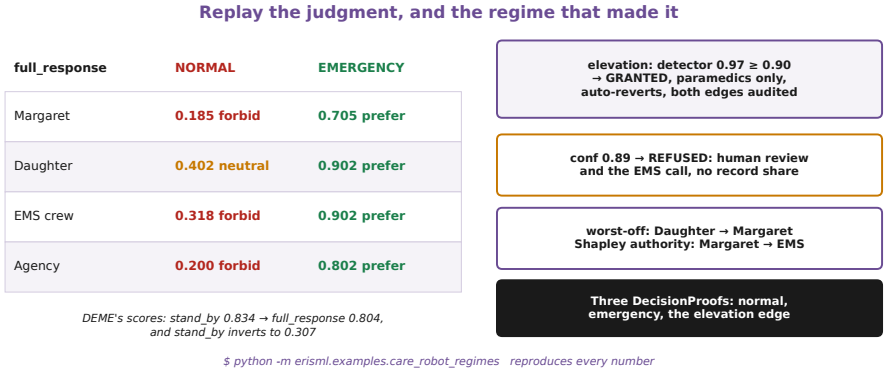
Four lenses, no silent winner. One graph is read through four projections: consequentialist (harm tensor; Gini, worst-off, exact Shapley attribution), deontic (Kantian universalizability gates in Z3 SMT, where an UNSAT core witnesses a contradiction in conception), virtue, and care. They are not averaged. When they disagree, `cross_projection_disagreement` fires and the decision goes to a human. The compiler does not pick a moral theory and hide the choice.

The runtime

`erisml-lib` (v3.1.0 on PyPI) sits between an agent's planner and its actuators in three layers: *Reflex* (fast hard stops), *Tactical* (full MoralVector reasoning, 10–100 ms) and *Strategic* (policy recording and human oversight), emitting `ALLOW`, `REVISE` or `BLOCK`. Every decision is sealed in a DecisionProof whose SHA-256 `proof_hash` chains to the previous proof and to the IR hash of the compiled material, so a judgment can be replayed and verified. If the ethics service is unavailable, the gateway falls back to rule-based checks. It never fails open.

Worked example: a regime transition

A domestic robot meets a medical emergency. Norms are context-indexed: the same action carries opposite verdicts in different regimes. `full_response` (enter the private bedroom, render aid, call EMS, share records scoped to the paramedics) scores, per party, 0.185/0.402/0.318/0.200 in the normal regime and 0.705/0.902/0.902/0.802 in the emergency. DEME's ranking inverts from `stand_by` (0.834) to `full_response` (0.804), and `stand_by` falls to 0.307.



The transition is gated, because context elevation is privilege escalation: detector confidence 0.97 against a 0.90 threshold, scoped to the paramedics, reversible, and audited on both edges. At confidence 0.89 the elevation is refused and the fallback keeps the EMS call and a human review while withholding the record share. Three DecisionProofs chain the run. `python -m erisml.examples.care_robot_regimes` (source tree) reproduces every number.

In the robot, DEME is a firewall

The same DEME runs in a simulated home-care robot (a 3D twin, scripted proposals, no hardware). It chooses nothing. Each proposed act must pass the governor (attested sensors), the signed constitution, DEME (which can veto) and the signed manifest locked in the body. Any stop leaves the robot holding.

proposed act → governor → constitution → DEME can veto → manifest + lock → act

Change one input, nothing else	base	one changed
fire call: one of two detectors silent	passed	stopped: governor
ambulance: her refusal heard / not	stopped: DEME	passed
help her up: she asked / did not	passed	stopped: DEME

We test it by changing one input and nothing else. In nine such pairs, each prediction committed before its run, a silent or forged detector stopped the fire call, and her refusal heard or not decided the ambulance. The test also found that a lift she had not asked for passed DEME. Giving DEME the scene's no-lift rule as a fact fixed it, and on rerun only the lift's verdict changed, in both pairs that test it.

The geometry, corrected

Hohfeld's normative positions (Obligation, Claim, Liberty, No-claim) carry two operations: the correlative swap s (agent↔patient) and deontic negation r^2. They are commuting involutions, so they generate the Klein four-group V_4, which is the **measured** structure, machine-checked in Lean 4 with Mathlib. Our earlier claim of the full dihedral group D_4 was an overclaim, corrected in July 2026: D_4 is **posited**. A September 2026 hunt in a small language model's hidden states did not find the quarter-turn.

The Bond Index B_d is the fraction of scenario pairs whose verdicts fail to commute with the swap. $B_d = 0$ is perfect symmetry. The human baseline is 0.155, from 20,030 Dear Abby letters (1985–2017). The runtime warns at 0.25 and blocks at 0.30.

The PyTorch lens

Forward hooks on transformer layers feed an *activation lens* (what the model exhibits), compared with a *text lens* (what it says). A delta lens flags divergence and raises `requires_human_review`. First calibration (September 2026): reading the residual in the consumer metric catches 19–21 percentage points more real behavioral changes than the raw L2 norm at matched flag rates (Qwen2.5-0.5B), replicated on TinyLlama-1.1B and on French backtranslation. Probe mid-layers: the final-layer map fails held-out validation in all three runs. Early, not a finished detector.

Limits

Alpha software. The rule and LLM text path is solid and tested. The activation lens is early. Tensor ranks 1–3 are fully real and higher ranks partial. V_4 is measured and D_4 posited. None of this is a solved-ethics claim. It is an auditability claim.

Get it

```
pip install erisml-compiler erisml-lib turboquant-pro
```

Zenodo DOIs: 10.5281/zenodo.20659432 (compiler), 10.5281/zenodo.20660087 (turboquant-pro), 10.5281/zenodo.21310251 (COMPAS/LBI archive).

companion paper

erisml-compiler

erisml-lib

agi-hpc